

Entanglement entropy

Density matrices, field theory, and holography

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Entanglement entropy connects quantum information with field-theory scaling and gravitational geometry. Its definition is simple for a finite bipartite system, but its interpretation becomes more delicate for mixed states, continuum fields, and gauge theories.

This note begins with explicit density-matrix calculations, develops replica methods in two-dimensional conformal field theory, and then derives basic holographic examples. The final sections explain modular Hamiltonians, quantum corrections, and reconstruction.

The prerequisites are linear algebra, quantum mechanics, and elementary field theory. Natural logarithms are used, so entropy is measured in nats. Units are $\hbar = k_B = 1$. The notation for density matrices, subsystem labels, and entropy follows the standard conventions of the cited papers.

These are pedagogical study notes, with worked calculations and reading references. They summarize established results and indicate the assumptions behind semiclassical or conjectural statements.

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1 Reduced states and what entropy measures

For a finite system with a tensor-product Hilbert space, the reduced state of A is

$$\rho_A = \text{Tr}_B \rho_{AB}. \quad (1.1)$$

It reproduces every expectation value of an observable acting only on A . The von Neumann entropy is

$$S(A) = -\text{Tr} \rho_A \log \rho_A. \quad (1.2)$$

Diagonalizing ρ_A reduces this to $-\sum_i p_i \log p_i$, with $0 \log 0 = 0$. Positivity and unit trace imply nonnegative entropy.

If the global state is pure, $S(A)$ is a measure of bipartite entanglement between A and B . If the global state is mixed, $S(A)$ includes uncertainty that need not arise from entanglement. A subsystem of a product of two thermal states has positive entropy even though the two systems are unentangled.

Unitary conjugation of the full density matrix preserves its eigenvalues and therefore its entropy. A subsystem entropy can nevertheless change because interactions redistribute correlations between the subsystem and its complement. This is how entanglement growth is compatible with exactly unitary time evolution.

The tensor-product assumption is straightforward for a lattice of spins. In continuum quantum field theory, ultraviolet divergences and operator-algebra subtleties enter. In gauge theories, constraints can obstruct naive spatial factorization. These issues are addressed later rather than silently ignored.

Check. Entropy is a property of a state together with a choice of subsystem or observable algebra. It is not a number attached to a wavefunction without specifying what is retained.

2 Schmidt decomposition and a two-qubit calculation

A pure bipartite state admits a Schmidt decomposition,

$$|\psi\rangle = \sum_i \sqrt{p_i} |i\rangle_A |i\rangle_B. \quad (2.1)$$

Tracing out either side gives the same nonzero spectrum. Therefore

$$S(A) = S(B), \quad S(AB) = 0. \quad (2.2)$$

The two subsystem Hilbert spaces need not have equal dimensions; only their nonzero reduced eigenvalues agree.

Consider

$$|\psi\rangle = \sqrt{p}|00\rangle + \sqrt{1-p}|11\rangle. \quad (2.3)$$

The reduced density matrix is diagonal with entries p and $1-p$, so

$$S(A) = -p \log p - (1-p) \log(1-p). \quad (2.4)$$

The entropy vanishes at the product-state endpoints and is maximal at $p = 1/2$, where it equals $\log 2$.

Compare this pure state with the mixed state $p|00\rangle\langle 00| + (1-p)|11\rangle\langle 11|$. Both have exactly the same reduced states. The mixed state is separable and contains classical correlations, while the pure state is entangled for $0 < p < 1$.

This example establishes an important operational limit: measurements on A alone cannot determine whether its mixedness arises from classical preparation uncertainty or entanglement with B . One needs joint information or additional assumptions about the global state.

Check. At $p = 1/2$, the Bell state has global entropy zero, whereas the correlated classical mixture has global entropy $\log 2$. Their subsystem entropies are identical.

3 Renyi entropies and the entanglement spectrum

The Renyi entropies are

$$S_n(A) = \frac{1}{1-n} \log \text{Tr } \rho_A^n, \quad n > 0, n \neq 1. \quad (3.1)$$

Taking $n \rightarrow 1$ gives the von Neumann entropy, provided the limit is controlled. The second Renyi entropy is $-\log \text{Tr } \rho_A^2$, so it is determined by the purity.

For the two-qubit example,

$$S_n = \frac{1}{1-n} \log[p^n + (1-p)^n]. \quad (3.2)$$

At equal probabilities every Renyi entropy equals $\log 2$. Away from that point, the family resolves information that one entropy alone cannot capture.

The limit $n \rightarrow \infty$ gives $-\log p_{\max}$, while $n \rightarrow 0$ counts the logarithm of the rank when the support is finite. Thus different values of n emphasize different parts of the spectrum. The full collection of eigenvalues, often encoded through $-\log \rho_A$, is called the entanglement spectrum.

Knowing a few integer moments does not uniquely and stably reconstruct a continuum spectrum. In replica calculations, analytic continuation from integer n to a neighborhood of one is an additional step with assumptions. It should not be replaced by an unexamined formal substitution.

Renyi entropies generally do not obey every inequality satisfied by von Neumann entropy. When using an entropy inequality in field theory or gravity, verify which entropy is actually being computed.

Check. For a pure reduced state, $\text{Tr} \rho_A^n = 1$ for every positive n , so all Renyi entropies vanish.

4 Mutual information and entropy inequalities

The mutual information is

$$I(A : B) = S(A) + S(B) - S(AB). \quad (4.1)$$

It measures total correlations, including classical correlations and entanglement. Subadditivity gives $I(A : B) \geq 0$. In a finite tensor-product system it vanishes precisely when $\rho_{AB} = \rho_A \otimes \rho_B$.

For a pure bipartite state, $I(A : B) = 2S(A)$. For the equally weighted classical mixture in the previous example, it is only $\log 2$. Thus mutual information distinguishes these two joint states even though their separate reduced states agree.

Strong subadditivity can be written

$$S(AB) + S(BC) \geq S(B) + S(ABC). \quad (4.2)$$

Equivalently, conditional mutual information is nonnegative. It constrains how correlations can be shared and is stronger than ordinary subadditivity.

The Araki–Lieb inequality gives

$$|S(A) - S(B)| \leq S(AB). \quad (4.3)$$

For a pure global state this forces equality of the subsystem entropies. Conditional entropy $S(AB) - S(B)$ can be negative in quantum mechanics; a Bell pair gives $-\log 2$. This is impossible for a classical joint probability distribution.

In continuum field theory, individual spatial entropies are divergent. For separated regions, mutual information often cancels the local boundary divergences and remains finite. When boundaries approach or touch, additional divergences can arise. Regulator independence must be checked in the precise geometric limit.

Check. A statement about entanglement cannot be inferred solely from positive mutual information: a separable classical mixture already has nonzero mutual information.

5 Relative entropy and distinguishability

For two density matrices with compatible supports,

$$S(\rho\|\sigma) = \text{Tr } \rho(\log \rho - \log \sigma). \quad (5.1)$$

It is nonnegative and vanishes only when the states agree. It is not symmetric under exchanging its arguments. If ρ has support outside that of σ , the relative entropy can diverge.

Partial trace cannot increase distinguishability:

$$S(\rho_{AB}\|\sigma_{AB}) \geq S(\rho_A\|\sigma_A). \quad (5.2)$$

This data-processing property provides a powerful organizing principle for entropy inequalities and for restrictions on physical channels.

For a thermal reference state $\sigma = e^{-\beta H}/Z$,

$$S(\rho\|\sigma) = \beta \text{Tr}(\rho H) - S(\rho) + \log Z. \quad (5.3)$$

Subtracting the same expression evaluated in σ gives

$$S(\rho||\sigma) = \beta\Delta\langle H \rangle - \Delta S. \quad (5.4)$$

Positivity therefore says that the thermal state minimizes the appropriate free-energy functional at fixed Hamiltonian and temperature.

For diagonal density matrices, the expression reduces to classical relative entropy between probability distributions. Quantum noncommutativity adds structure, but the basic interpretation as distinguishability survives.

Check. An entropy difference by itself can be negative. Relative entropy remains nonnegative because it includes the change in expectation value of the reference modular Hamiltonian.

6 Modular Hamiltonians and the first law

The modular Hamiltonian of a reference state is conventionally written

$$K = -\log \sigma. \quad (6.1)$$

Its additive normalization is fixed by $\text{Tr } e^{-K} = 1$. It is generally nonlocal and should not be identified with the ordinary physical Hamiltonian.

For a small trace-preserving perturbation about σ , differentiating the entropy gives

$$\delta S = \text{Tr}(\delta\rho K) = \delta\langle K \rangle. \quad (6.2)$$

This is the first law of entanglement. It is a linearized identity around a fixed reference state, not an exact relation for finite perturbations.

Relative entropy makes the correction transparent:

$$S(\rho||\sigma) = \Delta\langle K \rangle - \Delta S \geq 0. \quad (6.3)$$

The first-order term vanishes at the reference state because relative entropy has a minimum there. The leading nonzero term is typically quadratic and encodes a positive information metric under suitable regularity assumptions.

For the vacuum of a relativistic theory restricted to a half-space, the modular generator is a weighted boost energy. At time zero, with the region $x^1 > 0$,

$$K = 2\pi \int_{x^1 > 0} d^{d-1}x \, x^1 T_{00} + \text{constant}. \quad (6.4)$$

The position-dependent weight is why a modular Hamiltonian can be local in a special geometry while still differing from ordinary time translation.

Check. In applying the first law, hold the reference K fixed. Replacing it by the modular Hamiltonian of the perturbed state changes the variation being computed.

7 A ball in a conformal vacuum

Conformal symmetry maps the vacuum modular flow of a half-space to that of a ball. For a ball of radius R on a constant-time slice of a d -dimensional CFT,

$$K = 2\pi \int_{r < R} d^{d-1}x \frac{R^2 - r^2}{2R} T_{00} + \text{constant}. \quad (7.1)$$

The weight vanishes at the entangling surface. This is a special local formula; a generic region has a nonlocal modular Hamiltonian.

For a perturbation whose energy density is approximately constant across a small ball, the first law gives

$$\delta S = \frac{2\pi\Omega_{d-2}R^d}{d^2 - 1}, \delta\langle T_{00} \rangle. \quad (7.2)$$

To verify the coefficient, perform the radial integral:

$$\frac{1}{2R} \int_0^R dr r^{d-2} (R^2 - r^2) = \frac{R^d}{(d-1)(d+1)}. \quad (7.3)$$

Multiplying by $2\pi\Omega_{d-2}$ gives the displayed result.

The relation is an expansion in the small perturbation and, for a nonuniform state, in gradients across the ball. It does not assign a single physical temperature to all points in the region. The modular weight is spatially varying.

This calculation is useful in holography because the corresponding first-order change in minimal area is constrained by the bulk metric perturbation. With suitable assumptions, demanding the boundary first law for many balls constrains bulk gravitational equations.

Reading connection: The conformal mapping of ball entanglement to a thermal problem is developed by Casini, Huerta, and Myers [4].

8 Why field-theory entropy is ultraviolet divergent

A local quantum field has fluctuations at arbitrarily short distances. Modes on opposite sides of a spatial boundary are correlated at every scale. A lattice regulator cuts off the shortest correlations and renders the reduced density matrix well-defined in an ordinary tensor-product setting.

For a smooth entangling surface in a d -dimensional spacetime with $d > 2$, the leading vacuum divergence generally scales as

$$S \sim \frac{\text{Area}(\partial A)}{\epsilon^{d-2}} \times \text{regulator-dependent coefficient}. \quad (8.1)$$

The exponent follows because the entangling surface has $d - 2$ spatial dimensions. Subleading curvature terms and universal logarithms can contain physical information that survives changes in the short-distance regulator.

In two spacetime dimensions, the entangling surface consists of endpoints and the vacuum CFT entropy grows logarithmically with interval size. At finite temperature, a sufficiently large region also has a thermal volume contribution. The area law is therefore not a universal description of every state or every scale.

Continuum local algebras are typically type III, so the naive finite-dimensional density-matrix language does not transfer literally without a regulator. Relative entropy and algebraic formulations can remain meaningful where an individual von Neumann entropy diverges.

Gauge constraints introduce an additional issue beyond ultraviolet divergence. A definition of the region's observable algebra, treatment of boundary electric flux, or extension of the Hilbert space must be specified. Different choices can differ by boundary contributions.

Check. A divergent entropy is not automatically a failure of the theory. Ask which differences, universal coefficients, or correlation measures remain well-defined.

9 The replica construction

A Euclidean path integral prepares a wavefunctional. To form a reduced density matrix, glue the bra and ket fields along the complement of the retained region while leaving a cut along that region. Multiplying n copies and taking the trace cyclically glues the cuts.

For integer n , this constructs an n -sheeted geometry and gives

$$\mathrm{Tr} \rho_A^n = \frac{Z_n}{Z_1^n}. \quad (9.1)$$

The normalization by Z_1^n is essential. The entropy is obtained from

$$S(A) = - \partial_n \log \frac{Z_n}{Z_1^n} \Big|_{n=1}. \quad (9.2)$$

The calculation thus replaces a spectral problem for a reduced density matrix with a partition function on a branched geometry.

The replica construction is exact at integer n when the path integral and regulator are well-defined. Analytic continuation to noninteger n , saddle selection, and interchange of limits are separate steps. In a gravitational calculation, there may be several replica saddles whose dominance changes with parameters.

One should also distinguish $\log \overline{Z_n}$ from $\overline{\log Z_n}$ in theories with disorder or ensemble averaging. Different averaging procedures compute different quantities and can matter for entropy interpretations.

Check. At $n = 1$, $Z_n/Z_1^n = 1$. An unnormalized replica answer that fails this elementary check cannot be inserted directly into the entropy derivative.

Reading connection: For CFT applications, see Calabrese and Cardy [2,3].

10 Twist fields and one interval in a CFT

In a two-dimensional CFT, the branch points of the replica construction can be represented by twist fields in the replicated theory. For an interval with endpoints separated by ℓ , the relevant two-point function is fixed by conformal symmetry.

The twist field has holomorphic and antiholomorphic weights

$$h_n = \bar{h}_n = \frac{c}{24} \left(n - \frac{1}{n} \right). \quad (10.1)$$

Its full scaling dimension is their sum. Therefore

$$\text{Tr } \rho_A^n = c_n \left(\frac{\ell}{\epsilon} \right)^{-\frac{c}{6}(n-1/n)}. \quad (10.2)$$

The coefficient c_n depends on the regulator and normalization. Taking the logarithm and dividing by $1 - n$ gives

$$S_n = \frac{c}{6} \left(1 + \frac{1}{n} \right) \log \frac{\ell}{\epsilon} + \text{constant}_n. \quad (10.3)$$

The von Neumann limit is

$$S = \frac{c}{3} \log \frac{\ell}{\epsilon} + \text{constant}. \quad (10.4)$$

The central charge fixes the universal logarithmic slope, while the additive constant is not universal.

The twist dimensions can be derived by uniformizing the branched surface and evaluating the Schwarzian term in the stress-tensor transformation. The factor from summing over the n replicas is crucial; forgetting it changes $n - 1/n$ into an incorrect expression.

Check. The twist's full dimension is $2h_n$. The two-point function decays with twice that full dimension. Keeping those two factors separate avoids the common factor-of-two mistake.

11 Finite temperature and a finite circle

A conformal map from the plane to a cylinder replaces the vacuum interval distance by a hyperbolic sine for a thermal state on an infinite line:

$$S = \frac{c}{3} \log \left[\frac{\beta}{\pi \epsilon} \sinh \left(\frac{\pi \ell}{\beta} \right) \right] + \text{constant}. \quad (11.1)$$

For $\ell \ll \beta$, expanding the hyperbolic sine recovers the vacuum logarithm plus a correction

$$S - S_{\text{vac}} = \frac{c\pi^2 \ell^2}{18\beta^2} + O(\ell^4/\beta^4). \quad (11.2)$$

For $\ell \gg \beta$, the leading term is $\pi c \ell / (3\beta)$, matching the thermal entropy density times the interval length.

For the vacuum on a spatial circle of circumference L ,

$$S = \frac{c}{3} \log \left[\frac{L}{\pi \epsilon} \sin \left(\frac{\pi \ell}{L} \right) \right] + \text{constant}. \quad (11.3)$$

The equality under $\ell \leftrightarrow L - \ell$ is the expected purity relation. These two formulas concern different geometries and states; replacing sine by hyperbolic sine without tracking the Euclidean identifications hides that distinction.

A theory on a circle at nonzero temperature has a torus path integral. A single elementary conformal map does not determine its general interval entropy solely from c . Additional spectral and dynamical information is needed.

Check. In the short-interval limit both expressions approach the same vacuum logarithm. In the long-interval regime only the thermal infinite-line case develops a thermal volume term.

12 Gaussian states and a reproducible lattice method

For a bosonic quadratic Hamiltonian with canonical coordinates and momenta, a Gaussian state is determined by its covariance matrix. When the mixed coordinate-momentum covariance vanishes, the reduced state of a set of sites can be characterized by the restricted matrices

$$X_{ij} = \langle q_i q_j \rangle, \quad P_{ij} = \langle p_i p_j \rangle. \quad (12.1)$$

The positive symplectic eigenvalues are the square roots of the eigenvalues of $X_A P_A$. For a physical state they are at least $1/2$ in the convention $[q_i, p_j] = i\delta_{ij}$.

The entropy is

$$S(A) = \sum_j \left[\left(\nu_j + \frac{1}{2} \right) \log \left(\nu_j + \frac{1}{2} \right) \right. \quad (12.2)$$

$$\left. - \left(\nu_j - \frac{1}{2} \right) \log \left(\nu_j - \frac{1}{2} \right) \right]. \quad (12.3)$$

Each symplectic mode behaves like an effective thermal oscillator. A pure mode has $\nu_j = 1/2$ and contributes zero entropy.

For $H = (p^T p + q^T K q)/2$ with a positive stiffness matrix, the ground-state covariances are $X = K^{-1/2}/2$ and $P = K^{1/2}/2$. Restrict the covariances only after computing them for the full system. Diagonalizing the isolated subsystem Hamiltonian would compute a different state.

In numerical work, a symmetric matrix similar to $X_A P_A$ is preferable to multiplying nonsymmetric matrices directly. Zero modes require an infrared regulator or a careful treatment of boundary conditions. A massless periodic chain without such a prescription has an ill-defined oscillator zero-mode ground state.

Check. For the complete pure system, all symplectic eigenvalues are $1/2$. This is a stringent normalization check before computing any subsystem entropy.

13 The classical holographic prescription

In a static holographic state with a leading Einstein-gravity description,

$$S(A) = \frac{\text{Area}(\gamma_A)}{4G} \quad (13.1)$$

where γ_A is a minimal surface anchored on ∂A and homologous to A . It has codimension two in spacetime. The homology constraint rules out geometrically short surfaces that do not represent the correct entropy problem.

For a vacuum interval in AdS_3 , the constant-time metric is $L^2(dx^2 + dz^2)/z^2$. A semicircle $x^2 + z^2 = (\ell/2)^2$ is the extremal curve. Its regulated length is $2L \log(\ell/\epsilon)$ to leading order.

Using the Brown–Henneaux relation gives

$$\frac{2L}{4G_3} \log \frac{\ell}{\epsilon} = \frac{c}{3} \log \frac{\ell}{\epsilon}. \quad (13.2)$$

This reproduces the universal CFT coefficient without inserting a density matrix into the bulk calculation directly.

The area prescription computes the leading large-central-charge contribution in its domain of validity. Quantum bulk fields contribute at lower

orders, and higher-derivative gravitational actions generally require a modified geometric functional.

The derivation through gravitational replicas explains why extremality and homology arise. It does not imply that every arbitrary spacetime with a minimal surface defines a consistent boundary quantum theory.

Reading connection: The original proposal is Ryu and Takayanagi [5]; the gravitational replica argument is Lewkowycz and Maldacena [7].

14 Two intervals and competing surfaces

Take two equal intervals of length ℓ separated by a gap h on an infinite line. At leading classical order in a holographic vacuum, two surface pairings compete. Their entropies are

$$S_{\text{disc}} = \frac{2c}{3} \log \frac{\ell}{\epsilon}, \quad (14.1)$$

$$S_{\text{conn}} = \frac{c}{3} \log \frac{h(2\ell + h)}{\epsilon^2}. \quad (14.2)$$

Choosing the smaller value gives

$$I(A : B) = \frac{c}{3} \max \left\{ 0, \log \frac{\ell^2}{h(2\ell + h)} \right\}. \quad (14.3)$$

The transition occurs at $h/\ell = \sqrt{2} - 1$.

The vanishing of the leading mutual information in the disconnected phase does not mean the exact state factorizes. Bulk quantum corrections can give a nonzero contribution of order one even when the order- c term is zero. The sharp classical transition is smoothed or corrected in a finite-parameter theory.

In a general CFT, the entropy of two intervals depends on a twist-field four-point function and therefore on more than the central charge. The simple minimum formula is a special holographic large- c result under the relevant spectrum and saddle assumptions.

This example illustrates why entropy calculations can change qualitatively without a singular local geometry. The change is in which admissible surface dominates an optimization problem.

Check. The cutoff cancels from the mutual information for separated intervals. If a residual $\log \epsilon$ survives in this calculation, one of the endpoint contributions has been counted incorrectly.

15 A strip in higher-dimensional AdS

For a static strip of width ℓ and large transverse area $V_\perp$ in a d -dimensional boundary theory, describe the surface by $z(x)$. Its area is

$$\text{Area} = L^{d-1} V_\perp \int dx z^{-(d-1)} \sqrt{1 + z'^2}. \quad (15.1)$$

Translation invariance in x gives a first integral. At the turning point $z = z_*$, $z' = 0$, so

$$\frac{z^{-(d-1)}}{\sqrt{1 + z'^2}} = z_*^{-(d-1)}. \quad (15.2)$$

Consequently,

$$\frac{\ell}{2} = z_* \int_0^1 du \frac{u^{d-1}}{\sqrt{1 - u^{2(d-1)}}}. \quad (15.3)$$

The turning depth is proportional to the strip width, demonstrating explicitly the relation between bulk depth and boundary length scale.

For $d > 2$, the regulated area has the structure

$$\text{Area} = \frac{2L^{d-1}V_\perp}{(d-2)\epsilon^{d-2}} - \frac{\text{positive constant } L^{d-1}V_\perp}{\ell^{d-2}} + \dots. \quad (15.4)$$

The divergent term is local at the two boundary components. The finite width dependence is fixed by scaling, with a coefficient obtained from the integral.

The case $d = 2$ must be taken separately: the power divergence becomes a logarithm. Substituting $d = 2$ directly into a formula containing $1/(d-2)$ without regulating the limit obscures that transition.

Check. There are two branches of the strip surface. The factor of two in both width and area calculations is geometric, not a convention for entropy.

16 Time dependence and extremal surfaces

A Lorentzian surface cannot be selected by minimizing area over arbitrary spacetime deformations: timelike variations make such a prescription ill-posed. The covariant holographic proposal instead uses an extremal surface with the appropriate anchoring and homology conditions, followed by the correct comparison among candidates.

For a static state, this reduces to the minimal surface on a time-reflection-symmetric slice. In time-dependent geometries, the extremal surface generally leaves the boundary time slice. This is why a sequence of instantaneous spatial minima need not compute the entropy correctly.

After a global quench, entanglement can grow even while total entropy remains zero. In integrable models, quasiparticle pictures can explain characteristic growth by tracking entangled pairs. In strongly interacting holographic settings, geometric growth and saturation can be analyzed through evolving extremal surfaces. These are different dynamical descriptions and should not be assumed to yield identical detailed behavior.

An entanglement growth velocity is also distinct from a butterfly velocity or a microscopic causal velocity. Inequalities between them require specified assumptions about dimension, dynamics, and the state.

The causal domain of a boundary region and its entanglement wedge need not coincide. Entanglement can encode information deeper in the bulk than would be accessible through a simple causal-wedge construction.

Reading connection: The covariant proposal is Hubeny, Rangamani, and Takayanagi [6]. The relation between entanglement and scrambling is discussed further in the quantum-chaos note.

17 Quantum corrections and generalized entropy

At the first quantum order around a classical bulk geometry, the entropy includes bulk entanglement across the classical surface, together with the appropriate local renormalization terms. A useful schematic expression is

$$S(A) = \frac{\text{Area}(\gamma_A)}{4G} + S_{\text{bulk}} + \cdots \quad (17.1)$$

The first term is typically of order the boundary central charge, while the entropy of a fixed number of bulk fields is order one.

Beyond the strictly leading surface position, the generalized entropy is extremized. This gives a quantum extremal surface rather than an area extremum alone. The quantum correction can shift both the surface and its value, with the perturbative order depending on the expansion being performed.

Neither the area term nor bulk entropy should be renormalized independently with incompatible cutoffs. Their divergences are linked through the effective gravitational action. Higher-curvature interactions also modify the geometric entropy functional.

Competing quantum extremal surfaces are central to island calculations of radiation entropy. The island contribution changes which bulk degrees of freedom enter the matter entropy. It does not mean that entropy is created or destroyed by drawing an arbitrary new boundary.

Check. “Add bulk entropy to RT” and “extremize generalized entropy” agree at the appropriate first perturbative order but are not identical prescriptions at all orders. State the accuracy being claimed.

Reading connection: The leading quantum correction is [8]; quantum extremal surfaces are [9]; higher-derivative functionals are discussed in [11].

18 Relative entropy and bulk reconstruction

Within a suitable semiclassical code subspace, boundary relative entropy can match bulk relative entropy in the corresponding entanglement wedge, to the order controlled by the gravitational expansion. The area contribution appears in modular quantities and cancels appropriately in the comparison.

This relation gives a precise sense in which a boundary subregion retains information about a bulk region. If two nearby code-subspace states are distinguishable in the bulk wedge, the boundary region can retain the corresponding distinguishability.

Quantum error correction provides the right interpretation of multiple reconstructions. Different boundary operators can represent the same logical bulk observable on a restricted subspace. They are not independent

copies of an arbitrary quantum state, so there is no violation of the no-cloning theorem.

The restriction to a code subspace matters. Large backreaction can change the geometry and the relevant wedge. A formula established for perturbative excitations about one background cannot simply be applied to an unrestricted collection of black-hole microstates.

Gauge constraints also matter for what counts as a bulk observable and how the subregion algebra is defined. Entanglement-wedge reconstruction is therefore more precise than the slogan that “all information inside a surface is stored on its boundary.” The algebra, state family, and accuracy must be supplied.

Reading connection: The relative-entropy relation is developed by Jafferis, Lewkowycz, Maldacena, and Suh [10].

19 Worked checks: thermal intervals and the first law

For a two-dimensional CFT on an infinite line, the thermal energy density above the vacuum is

$$\Delta\langle T_{00}\rangle = \frac{\pi c}{6\beta^2}. \quad (19.1)$$

For an interval $[-\ell/2, \ell/2]$, the vacuum modular Hamiltonian gives

$$\Delta\langle K\rangle = 2\pi\Delta\langle T_{00}\rangle \int_{-\ell/2}^{\ell/2} dx \frac{\ell^2/4 - x^2}{\ell}. \quad (19.2)$$

The integral equals $\ell^2/6$, so

$$\Delta\langle K\rangle = \frac{c\pi^2\ell^2}{18\beta^2}. \quad (19.3)$$

This matches the leading short-interval thermal entropy correction in Section 11.

The next term in $\log(\sinh x/x)$ is $-x^4/180$. Hence the thermal entropy difference is smaller than the modular-energy difference at the next order:

$$\Delta\langle K\rangle - \Delta S = \frac{c\pi^4\ell^4}{540\beta^4} + \cdots \geq 0. \quad (19.4)$$

This is a direct check of relative-entropy positivity. It also shows why the first law is only linearized: the higher-order correction is physically meaningful rather than an algebraic error.

The calculation assumes a small interval compared with the thermal scale. It should not be extrapolated to $\ell \gg \beta$, where the thermal volume term dominates and the perturbation of the reduced vacuum state is no longer small.

20 Worked checks: spectra and surface transitions

A binary spectrum. For $p = 1/4$, the von Neumann entropy is

$$S = -\frac{1}{4} \log \frac{1}{4} - \frac{3}{4} \log \frac{3}{4} \simeq 0.562335. \quad (20.1)$$

The second Renyi entropy is

$$S_2 = -\log \left(\frac{1}{16} + \frac{9}{16} \right) = \log \frac{8}{5} \simeq 0.470004. \quad (20.2)$$

The inequality $S_2 \leq S$ is a useful numerical check. At $p = 1/2$ they coincide because the nonzero spectrum is flat.

The two-interval transition. Equate the connected and disconnected surface entropies. Cutoff factors cancel, leaving $\ell^2 = h(2\ell + h)$. The positive root is $h/\ell = \sqrt{2} - 1$. For a much larger gap the disconnected saddle wins; for a much smaller gap the connected saddle wins.

An oscillator mode. For a symplectic eigenvalue $\nu = 1$, the entropy contribution is $(3/2) \log(3/2) - (1/2) \log(1/2)$. It is positive and vanishes continuously as $\nu \rightarrow 1/2$. Numerical routines should implement the vanishing $x \log x$ term by continuity to avoid spurious NaNs.

These examples test three distinct implementations: spectral entropy, geometric saddle comparison, and Gaussian covariance entropy. Agreement of their limiting behaviors is more informative than merely obtaining a nonnegative number from a numerical routine.

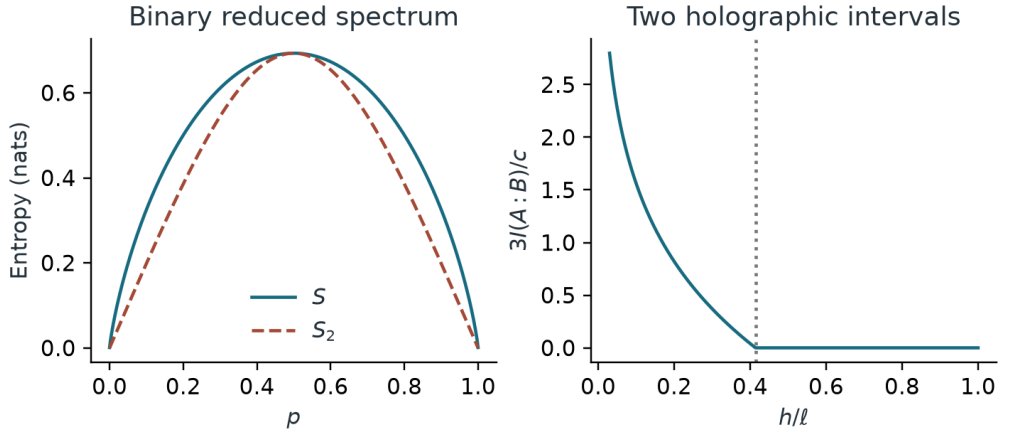


Figure 1: Analytic checks of spectral and holographic entropy. Left: von Neumann and second Renyi entropy for a binary reduced spectrum. Right: leading classical mutual information for equal separated intervals; quantum corrections need not vanish beyond the transition.

21 How to use entropy without overinterpreting it

Begin by specifying the global state, the subsystem or algebra, and the regulator. Decide whether the question concerns entanglement, total correlations, thermal uncertainty, or distinguishability. The same von Neumann formula can enter all four discussions, but its interpretation changes.

For a pure bipartite state, reduced entropy is an entanglement measure. For a mixed bipartite state, measures such as entanglement negativity or relative-entropy-based quantities address different questions and require separate definitions. Mutual information counts total correlations and does not isolate the quantum part.

For a CFT calculation, separate universal scale dependence from additive cutoff constants. For multiple intervals, do not assume central charge alone determines the answer. For a holographic calculation, state whether the result is classical, includes bulk quantum entropy, or uses a higher-derivative functional.

References and further reading

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